



Leicester, United Kingdom
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RANJAY KUMAR, R&D Engineer @Honeywell

- Skilled and motivated Engineer with 7+ years of experience in Embedded software development in linux and RTOS using C, C++ and python programming language. Proficient with tools like Git, JTAG debuggers, oscilloscopes, logic analyzers, and Linux environments

Employment History

Embedded Software Engineer at HONEYWELL INTERNATIONAL, INC., Leicester

October 2021 — Present

Projects:

- **SIA DC09 alarm reporting** integration as a python module.
- **MPI** : Ported an Intrusion Detection Application that was originally running on a microprocessor in a Linux environment to a cheaper micro controller (i. MXRT 1062) running on FreeRTOS.
- **HLD-Ubuntu upgrade** : Upgraded the operating system on the Panel with secure booting from HLD to Ubuntu (i. MXRT 1062).
- Carried on dealing with day to day feature development and bug fixes and support customer with issue resolution on live sites in Agile team.

Responsibilities:

- Firmware development for embedded systems used in Intrusion Detection Devices designed by Honeywell using C, C++ and python.
- Requirements scaling and designing framework enhancements.
- Information analysis to determine, recommend, and plan installation of new systems or modification of existing ones.
- Collaboration with teams in different domains to develop and deliver high quality product.

Achievements:

- Successfully completed projects within the stipulated time frame.
- Improved system functionality and efficiency by incorporating necessary upgrades.
- Honed my skills in programming, system design, and analysis along with new hardware while implementing new software solutions.
- Learned about new NXP hardware (i. MXRT 1052 and i. MXRT 1062) and other custom Honeywell board.
- Learned about various sensors being used including wireless sensor technology.
- Gained experience in various other micro controller and microprocessor systems.

Software Development Engineer at WATERS CORPORATION

November 2019 — Present

Roles:

- Software Development Lead Waters Corporation 11/2019 – 09/2021 Software
- Development and Support Waters Corporation 08/2018 – 11/2019

Responsibilities:

- Led the development and maintenance of Mass Lynx, a scientific instrument management application
 - Proficient in C and C++ languages and various libraries
 - Scaled requirements and designed software to support different instruments
 - Improved software performance by upgrading interfaces and correcting errors
 - Analysed information and collaborated with data processing and project managers
- Achievements:
- Successfully led development of Mass Lynx, contributing to organisation's success
 - Deep understanding of scaling requirements and designing software to support different instruments
 - Collaborated with data processing and project managers for successful projects.
 - Gained experience with better debugging skills.

Education

Bachelors of Technology in EEE, National Institute of Technology (NIT), Nagaland

May 2014 — November 2018

Patent/Publications

US11670145B1 : Exit route indication via synchronized audible cues, Leicester

December 2022

Skills

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|--|--|
| ☒ C ++ (Programming Language) | ☒ Data Structures & Algorithms |
| ☒ Serial Communications Protocols : SPI, I2C, UART, CAN, and Ethernet | ☒ Experience with Microprocessor & Microcontrollers |
| ☒ Agile Software Development | ☒ Python (Programming Language) |
| ☒ Debugging | ☒ Software Tools:- VS Studio, MCUXpresso, Eclipse, Atlassian Tools & Version Controlling, Oscilloscopes, JTAG, ect. |

Languages

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|---|---|
| ☒ English Highly proficient | ☒ Hindi Native speaker |
| ☒ Haryanvi Very good command | |

Hobbies

Cricket, Badminton, Table Tennis